7(a)	tungsten has a very high melting point	[1]
7(b)	Using $I / I_0 = e^{(-\mu x)}$ $0.5 = e^{(-2.46 \times x)}$ $x = \ln(2) / 2.46 = 0.282$ cm	[1] [1]
(c)(i)1	From Table 7.1, μ of muscle = 0.40 cm^{-1} Using $I / I_0 = e^{(-\mu x)}$ ratio along MM = $e^{(-0.40 \times 8.0)}$ = 0.041 or 0.0408	[1]
(c)(i)2	From Table 7.1, μ of bone = 2.46 cm^{-1} Using $I / I_0 = e^{(-\mu x)}$ For 2 cm of muscle, $I / I_0 = e^{(-0.40 \times 2.0)}$ or For 4 cm of muscle, $I / I_0 = e^{(-0.40 \times 4.0)}$ For 4 cm of bone, $I / I_0 = e^{(-2.46 \times 4.0)}$ Ratio along BB = $(I / I_0)_{muscle_2cm} \times (I / I_0)_{bone} \times (I / I_0)_{muscle_2cm}$ = $e^{(-0.40 \times 2.0)} \times e^{(-2.46 \times 4.0)} \times e^{(-0.40 \times 2.0)}$ = 0.0000108 or 0.000011 or Ratio = $e^{(-0.40 \times 4.0)} \times e^{(-2.46 \times 4.0)}$ = 0.0000108 or 0.000011	[1 - for each I / I_0] [1]

(c)(ii)	Since the ratio along MM is 0.041/0.000011 = 3727/ ≈ 10^3/ is about 3 orders of magnitude bigger/different than that along BB, the contrast is good.	[1] [1]
7(d)	To produce additional magnetic fields that cause the B to vary with position across patient This can be achieved when the magnetic field in the gradient coils superposed with the magnetic field from the main coil to produce varying magnetic fields at different positions.	[1] [1]
(e)	MRI is able to provide excellent contrast between the different parts of the brain (gray matter, white matter and blood) due to the distinct relaxation times in Table. 7.2, whereas X-ray has poor contrast between blood and brain due to similar absorption coefficients in Table 7.1	[1] [1]
(f)	(Any two relevant point) • small space, not suitable for patient who are claustrophobic • metal implants (e.g. pacemaker, cochlear implants) / ferromagnetic implants in patients may malfunction / move in the strong magnetic field • (long scan time) patient needs to remain still for long duration • any movement may cause blurred image, children or patients who cannot lie down for long duration and any reasonable challenges like loud noise which may be scary for some patients • high cost	[2]
(g)	For a solenoid, $B = \mu_0 n I$ $1.5 = (4\pi \times 10^{-7}) \times n \times 750$ $n = 1591.55\ m^{-1}$ number of turns = n × 2.2 = 3500 turns	 [1] [1]
(h)	Larmor frequency, $\omega = \gamma B_0$ $= (2.68 \times 10^8) \times 1.5 = 4.02 \times 10^8$ rad s^{-1} frequency of RF wave = $\omega / 2\pi$ $= 4.02 \times 10^8 / 2\pi$ $= 6.4 \times 10^7$ Hz	[1] [1]
(i)	Assuming all electrical energy is used to vaporise liquid helium: rate of electrical heating = rate of vaporisation of liquid helium $I^2 R = m_rate \times L_v$ (where m_rate is the initial rate of vaporization) $(750)^2(0.0045) = m_rate\ (21 \times 10^3)$ m_rate = 0.12 or 0.121 kg s^{-1}	 [1] [1]